Risk Factors, Predictors, and Outcomes of Post-Cardiopulmonary Bypass Vasoplegia: Regression and Machine Learning Approaches

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Background

Vasoplegia is a complication of cardiac surgery involving cardiopulmonary bypass (CPB) characterized by hypotension with low systemic vascular resistance despite preserved cardiac function.[1] It is associated with escalating vasoactive support, longer ICU stays, and increased mortality.[1,2] Incidence ranges widely (2.5 to 66.3%) across studies with a recent systematic review identifying 63 definitions in the literature.[2] CPB duration was the most consistent risk factor, likely reflecting inflammatory activation from prolonged blood-circuit contact,[1,3-5] but other modifiable intraoperative factors, such as hemoconcentration volume, remain understudied.[6] Existing prediction tools rely largely on preoperative or postoperative data, and whether minute-level intraoperative data available at CPB separation can identify at-risk patients earlier is unclear.

Objectives

1. Determine the institutional incidence of vasoplegia
2. Identify risk factors associated with vasoplegia
3. Evaluate the association between vasoplegia and postoperative outcomes
4. Describe the impact of vasoplegia specific therapies on vasoactive infusion requirements
5. Evaluate whether minute-level intraoperative data can predict vasoplegia at CPB separation

Methods

Design and Population

Retrospective single-center chart review of adults undergoing CPB (January 2025–May 2026). Of 728 patients screened, 24 were excluded for insufficient hemodynamic data (n=704). Incidence, regression analysis, and outcomes also excluded preoperative LVEF <50% and LVAD procedures (n=510).

Vasoplegia Definition & Classification

Vasoplegia was defined as mean arterial pressure (MAP) <65 mmHg during the first hour after CPB despite preserved cardiac function (Cardiac Index (CI) >2.2 L/min/m² or Left Ventricular Ejection Fraction >55%) and adequate preload (Central Venous Pressure > 8 mmHg). MAP <65 mmHg without these criteria were classified as hypotensive.

Data and Statistical Analysis

Data: Perioperative and postoperative variables were extracted from the electronic medical record.

Risk factors: Univariable logistic regression screening with significant variables (p<0.05) entered into a multivariable model.

Outcomes: Continuous outcomes compared via Mann-Whitney U test and categorical via Fisher's exact test.

Vasoplegia therapy: Descriptive analysis only (small subgroup sizes).

Predictive Models: Elastic net, random forest, and three-variable logistic models were compared by AUROC with nested 10-fold cross-validation.

Results

Classification	n (% of cohort)	95% CI
Vasoplegia	39 (7.6%)	5.49-10.31%
Hypotensive, vasoplegia criteria not met	14 (2.7%)	1.51-4.56%
No vasoplegia	457 (89.6%)	86.63-92.12%
Total	510 (100%)	

Table 1. Incidence of post-CPB vasoplegia

Surgery Type	No Vasoplegia (n=471)	Vasoplegia (n=39)
CABG	215 (45.6%)	7 (17.9%)
Valve Surgery	89 (18.9%)	6 (15.4%)
Combined CABG + Valve	53 (11.3%)	9 (23.1%)
Aorta	92 (19.5%)	13 (33.3%)
Other	22 (4.7%)	4 (10.3%)

Table 2. Surgical characteristics by vasoplegia status

Risk Factor	Unadjusted OR (95% CI)	p value	Adjusted OR (95% CI)	p value
CPB Duration	**1.01 (1.01-1.02)**	**<0.001**	**1.01 (1.00-1.02)**	**0.034**
Hemoconcentration Volume	**1.00 (1.00-1.00)**	**<0.001**	**1.00 (1.00-1.00)**	**0.022**
Intraoperative Norepinephrine, mcg	**1.00 (1.00-1.00)**	**<0.001**	**1.00 (1.00-1.00)**	**0.011**
Max Intraoperative Lactate	**1.39 (1.24-1.56)**	**<0.001**	**1.36 (1.13-1.67)**	**0.002**
Preoperative PT	1.22 (1.07-1.34)	0.002	1.62 (0.99-2.67)	0.054
Preoperative INR	13.4 (3.04-60.94)	<0.001	0.01 (0.00-5.91)	0.158
Autologous Blood, mL	1.00 (1.00-1.00)	0.021	1.00 (1.00-1.00)	0.256
PRBC, units	1.33 (1.15-1.52)	<0.001	0.81 (0.53-1.17)	0.310
Total Intraoperative Fluids	1.00 (1.00-1.00)	0.043	1.00 (1.00-1.00)	0.344
Preoperative Hematocrit	0.91 (0.87-0.96)	<0.001	0.85 (0.57-1.25)	0.407
FFP, Units	1.31 (1.13-1.50)	<0.001	0.86 (0.57-1.25)	0.464
Preoperative Hemoglobin	0.78 (0.67-0.89)	<0.001	1.37 (0.47-4.06)	0.561
Platelets, units	1.59 (1.23-2.03)	<0.001	1.15 (0.59-2.23)	0.681

Table 3. Risk factors for post-CPB vasoplegia

Bold = independent predictor after multivariable adjustment (p<0.05).
OR = odds ratio; PT = prothrombin time; INR = international normalized ratio; PRBC = packed red blood cell; FFP = fresh frozen plasma.

Outcome	No Vasoplegia (n=471)	Vasoplegia (n=39)	p value
Ventilator Duration, hours, median (IQR)	6.7 (3.70-14.08)	19.8 (7.49-49.91)	<0.001
ICU LOS, days, median (IQR)	2.08 (1.16-4.46)	5.50 (3.35-11.8)	<0.001
24 Hour Norepinephrine Equivalent, median (IQR)	0.02 (0.01-0.05)	0.07 (0.03-0.14)	<0.001
Early Mechanical Circulatory Support (48 hours), n (%)	19 (4.0)	2 (5.1)	0.670

Table 4. Postoperative outcomes by vasoplegia status

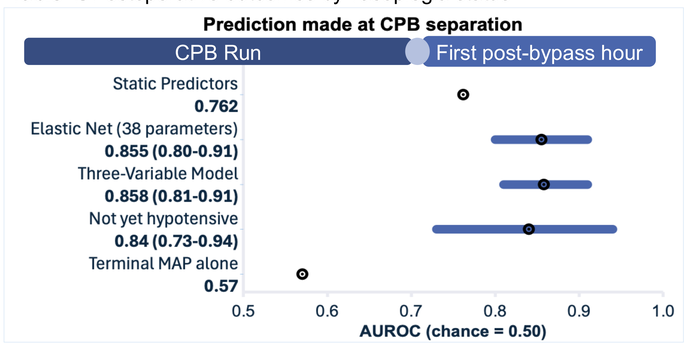


Figure 1. Prediction at CPB separation, AUROC (95% CI); n = 704, 45 events

Machine Learning Findings

- **Minute-level data improved discrimination:** AUROC 0.855 (elastic net) vs 0.762 (static predictors).
- **Three routine variables** (mean bypass MAP, CPB duration, hemoconcentration volume) reached AUROC 0.858, comparable to the 38-parameter model.
- **Not yet hypotensive at separation** (n=438, 16 events): AUROC 0.84 vs 0.57 for terminal MAP alone.

Results

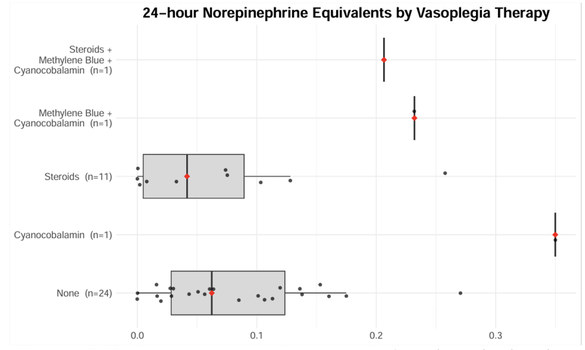


Figure 2. 24-hour norepinephrine equivalent (mcg/kg/min) by vasoplegia therapy received

Conclusions

- Vasoplegia occurred in 7.6% of patients, at the lower end of the reported incidence (2.5–66.3%).
- Vasoplegia was more common after combined CABG/valve procedures or aortic procedures consistent with greater surgical complexity among vasoplegia cases.
- Longer CPB duration, greater hemoconcentration volumes, higher intraoperative norepinephrine, and higher maximum intraoperative lactate independently predicted vasoplegia, identifying potential modifiable perioperative factors.
- Vasoplegia was associated with longer ventilation, longer ICU stays and vasoactive infusion requirements whereas early mechanical circulatory support use did not differ significantly between groups, though the small number of events limited analysis.
- In exploratory analysis, three routine variables available at CPB separation (mean bypass MAP, CPB duration, hemoconcentration volume) performed comparably to a 38-parameter model (AUROC 0.858 vs 0.855). Predictors were selected in the full cohort, and validation is needed.

Future Directions

- Expand the cohort to a 3-year retrospective review to improve analysis of uncommon outcomes including early mechanical circulatory support and to allow formal comparison of vasoplegia-specific therapies.
- Re-run predictive models on the updated cohort and repeat predictor selection within training folds.
- Validate a frozen model in an independent cohort toward real-time risk prediction at CPB separation.
- Further investigate hemoconcentration as a modifiable risk factors.

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